常用公式

1　三种坐标系常用关系

1.1　直角坐标系

直角坐标系如图 1 所示,单位矢量 $\boldsymbol{e}_x,\boldsymbol{e}_y,\boldsymbol{e}_z$ 满足右手螺旋关系。

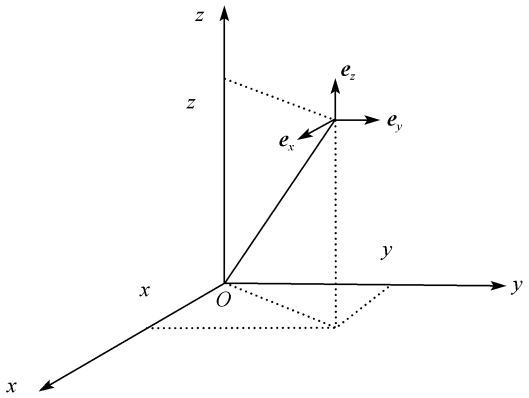


图 1　直角坐标系

直角坐标系中

坐标范围	$-\infty<x<\infty,-\infty<y<\infty,-\infty<z<\infty$
单位矢量	$\boldsymbol{e}_x,\boldsymbol{e}_y,\boldsymbol{e}_z$
单位矢量之间的关系	$\boldsymbol{e}_x\cdot\boldsymbol{e}_x=\boldsymbol{e}_y\cdot\boldsymbol{e}_y=\boldsymbol{e}_z\cdot\boldsymbol{e}_z=1,\boldsymbol{e}_x\cdot\boldsymbol{e}_y=\boldsymbol{e}_y\cdot\boldsymbol{e}_z=\boldsymbol{e}_z\cdot\boldsymbol{e}_x=0$ $\boldsymbol{e}_x\times\boldsymbol{e}_y=\boldsymbol{e}_z,\boldsymbol{e}_y\times\boldsymbol{e}_z=\boldsymbol{e}_x,\boldsymbol{e}_z\times\boldsymbol{e}_x=\boldsymbol{e}_y$
在直角坐标系中矢量表示	$\boldsymbol{a}(\boldsymbol{l})=\boldsymbol{a}(x,y,z)=\boldsymbol{e}_x a_x(x,y,z)+\boldsymbol{e}_y a_y(x,y,z)+\boldsymbol{e}_z a_z(x,y,z)$
微分长度元	$\mathrm{d}\boldsymbol{l}=\boldsymbol{e}_x\mathrm{d}x+\boldsymbol{e}_y\mathrm{d}y+\boldsymbol{e}_z\mathrm{d}z$
体积元	$\mathrm{d}V=\mathrm{d}x\mathrm{d}y\mathrm{d}z$
面积元	$\mathrm{d}S_x=\mathrm{d}y\mathrm{d}z,\mathrm{d}S_y=\mathrm{d}z\mathrm{d}x,\mathrm{d}S_z=\mathrm{d}x\mathrm{d}y,$ $\mathrm{d}\boldsymbol{S}=\boldsymbol{e}_x\mathrm{d}S_x+\boldsymbol{e}_y\mathrm{d}S_y+\boldsymbol{e}_z\mathrm{d}S_z$
哈密顿算符	$\nabla=\boldsymbol{e}_x\dfrac{\partial}{\partial x}+\boldsymbol{e}_y\dfrac{\partial}{\partial y}+\boldsymbol{e}_z\dfrac{\partial}{\partial z}$
梯度	$\nabla\varphi=\boldsymbol{e}_x\dfrac{\partial\varphi}{\partial x}+\boldsymbol{e}_y\dfrac{\partial\varphi}{\partial y}+\boldsymbol{e}_z\dfrac{\partial\varphi}{\partial z}$
散度	$\nabla\cdot\boldsymbol{a}=\dfrac{\partial a_x}{\partial x}+\dfrac{\partial a_y}{\partial y}+\dfrac{\partial a_z}{\partial z}$
旋度	$\nabla\times\boldsymbol{a}=\begin{vmatrix}\boldsymbol{e}_x & \boldsymbol{e}_y & \boldsymbol{e}_z\\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z}\\ a_x & a_y & a_z\end{vmatrix}$ $=\boldsymbol{e}_x\left(\dfrac{\partial a_z}{\partial y}-\dfrac{\partial a_y}{\partial z}\right)+\boldsymbol{e}_y\left(\dfrac{\partial a_x}{\partial z}-\dfrac{\partial a_z}{\partial x}\right)+\boldsymbol{e}_z\left(\dfrac{\partial a_y}{\partial x}-\dfrac{\partial a_x}{\partial y}\right)$
拉普拉斯运算	$\nabla^2\varphi=\Delta\varphi=\dfrac{\partial^2\varphi}{\partial x^2}+\dfrac{\partial^2\varphi}{\partial y^2}+\dfrac{\partial^2\varphi}{\partial z^2}$

1.2 圆柱坐标系

圆柱坐标系如图 2 所示,单位矢量 $\boldsymbol{e}_r$, $\boldsymbol{e}_\phi$,$\boldsymbol{e}_z$ 满足右手螺旋关系。

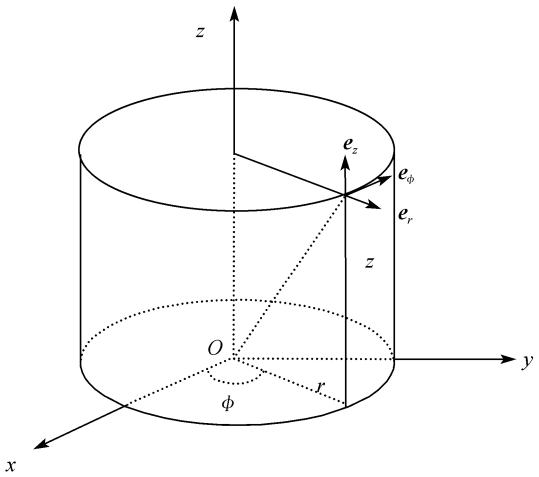


图 2　圆柱坐标系

圆柱坐标系中

坐标范围	$0\le r<\infty, 0\le \phi \le 2\pi, -\infty<z<\infty$
坐标变换关系	$r=\sqrt{x^2+y^2}, \tan\phi=\dfrac{y}{x}$
单位矢量	$\boldsymbol{e}_r, \boldsymbol{e}_\phi, \boldsymbol{e}_z$
单位矢量之间的关系	$\boldsymbol{e}_r\cdot\boldsymbol{e}_r=\boldsymbol{e}_\phi\cdot\boldsymbol{e}_\phi=\boldsymbol{e}_z\cdot\boldsymbol{e}_z=1, \boldsymbol{e}_r\cdot\boldsymbol{e}_\phi=\boldsymbol{e}_\phi\cdot\boldsymbol{e}_z=\boldsymbol{e}_z\cdot\boldsymbol{e}_r=0$ $\boldsymbol{e}_r\cdot\boldsymbol{e}_\phi=\boldsymbol{e}_z, \boldsymbol{e}_\phi\times\boldsymbol{e}_z=\boldsymbol{e}_r, \boldsymbol{e}_z\times\boldsymbol{e}_r=\boldsymbol{e}_\phi$
与直角坐标系单位矢量关系	$\boldsymbol{e}_r=\boldsymbol{e}_x\cos\phi+\boldsymbol{e}_y\sin\phi, \boldsymbol{e}_\phi=-\boldsymbol{e}_x\sin\phi+\boldsymbol{e}_y\cos\phi,$ $\boldsymbol{e}_x=\boldsymbol{e}_r\cos\phi-\boldsymbol{e}_\phi\sin\phi, \boldsymbol{e}_y=\boldsymbol{e}_r\sin\phi+\boldsymbol{e}_\phi\cos\phi$
在圆柱坐标系中矢量表示	$\boldsymbol{a}(\boldsymbol{l})=\boldsymbol{a}(r,\phi,z)=\boldsymbol{e}_r a_r(r,\phi,z)+\boldsymbol{e}_\phi a_\phi(r,\phi,z)+\boldsymbol{e}_z a_z(r,\phi,z)$
微分长度元	$\mathrm{d}\boldsymbol{l}=\boldsymbol{e}_r\mathrm{d}r+\boldsymbol{e}_\phi r\mathrm{d}\phi+\boldsymbol{e}_z\mathrm{d}z$
体积元	$\mathrm{d}V=r\mathrm{d}r\mathrm{d}\phi\mathrm{d}z$
面积元	$\mathrm{d}S_r=r\mathrm{d}\phi\mathrm{d}z, \mathrm{d}S_\phi=\mathrm{d}r\mathrm{d}z, \mathrm{d}S_z=r\mathrm{d}r\mathrm{d}\phi,$ $\mathrm{d}\boldsymbol{S}=\boldsymbol{e}_r\mathrm{d}S_r+\boldsymbol{e}_\phi\mathrm{d}S_\phi+\boldsymbol{e}_z\mathrm{d}S_z$
哈密顿算符	$\nabla=\boldsymbol{e}_r\dfrac{\partial}{\partial r}+\boldsymbol{e}_\phi\dfrac{1}{r}\dfrac{\partial}{\partial\phi}+\boldsymbol{e}_z\dfrac{\partial}{\partial z}$
梯度	$\nabla\varphi=\boldsymbol{e}_r\dfrac{\partial\varphi}{\partial r}+\dfrac{\boldsymbol{e}_\phi}{r}\dfrac{\partial\varphi}{\partial\phi}+\boldsymbol{e}_z\dfrac{\partial\varphi}{\partial z}$
散度	$\nabla\cdot\boldsymbol{a}=\dfrac{1}{r}\dfrac{\partial}{\partial r}(ra_r)+\dfrac{1}{r}\dfrac{\partial a_\phi}{\partial\phi}+\dfrac{\partial a_z}{\partial z}$
旋度	$\nabla\times\boldsymbol{a}=\begin{vmatrix}\frac{1}{r}\boldsymbol{e}_r & \boldsymbol{e}_\phi & \frac{1}{r}\boldsymbol{e}_z\\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\phi} & \frac{\partial}{\partial z}\\ a_r & ra_\phi & a_z\end{vmatrix}$ $=\boldsymbol{e}_r\left(\dfrac{1}{r}\dfrac{\partial a_z}{\partial\phi}-\dfrac{\partial a_\phi}{\partial z}\right)+\boldsymbol{e}_\phi\left(\dfrac{\partial a_r}{\partial z}-\dfrac{\partial a_z}{\partial r}\right)+\boldsymbol{e}_z\left(\dfrac{1}{r}\dfrac{\partial}{\partial r}(ra_\phi)-\dfrac{1}{r}\dfrac{\partial a_r}{\partial\phi}\right)$

拉普拉斯运算	$\nabla^2\varphi=\Delta\varphi=\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\varphi}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2\varphi}{\partial\phi^2}+\frac{\partial^2\varphi}{\partial z^2}$

1.3 球坐标系

球坐标系如图 3 所示,单位矢量 $\boldsymbol{e}_r,\boldsymbol{e}_\theta,\boldsymbol{e}_\phi$ 满足右手螺旋关系。

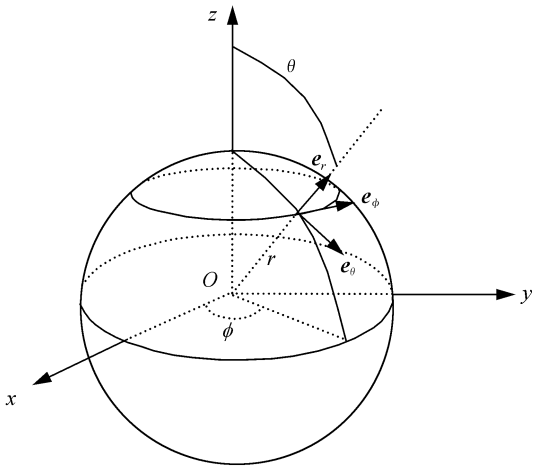


图 3 球坐标系

球坐标系中

坐标范围	$0\le r<\infty,0\le\theta\le\pi,0\le\phi<2\pi$
坐标变换关系	$r=\sqrt{x^2+y^2+z^2},\tan\theta=\frac{\sqrt{x^2+y^2}}{z},\tan\phi=\frac{y}{x}$
单位矢量	$\boldsymbol{e}_r,\boldsymbol{e}_\theta,\boldsymbol{e}_\phi$
单位矢量之间的关系	$\boldsymbol{e}_r\cdot\boldsymbol{e}_r=\boldsymbol{e}_\theta\cdot\boldsymbol{e}_\theta=\boldsymbol{e}_\phi\cdot\boldsymbol{e}_\phi=1,\boldsymbol{e}_r\cdot\boldsymbol{e}_\theta=\boldsymbol{e}_\theta\cdot\boldsymbol{e}_\phi=\boldsymbol{e}_\phi\cdot\boldsymbol{e}_r=0$ $\boldsymbol{e}_r\times\boldsymbol{e}_\theta=\boldsymbol{e}_\phi,\boldsymbol{e}_\theta\times\boldsymbol{e}_\phi=\boldsymbol{e}_r,\boldsymbol{e}_\phi\times\boldsymbol{e}_r=\boldsymbol{e}_\theta$
与直角坐标系单位矢量关系	$\boldsymbol{e}_r=\boldsymbol{e}_x\sin\theta\cos\phi+\boldsymbol{e}_y\sin\theta\sin\phi+\boldsymbol{e}_z\cos\theta,$ $\boldsymbol{e}_\theta=\boldsymbol{e}_x\cos\theta\cos\phi+\boldsymbol{e}_y\cos\theta\sin\phi-\boldsymbol{e}_z\sin\theta$ $\boldsymbol{e}_\phi=-\boldsymbol{e}_x\sin\phi+\boldsymbol{e}_y\cos\phi,$ $\boldsymbol{e}_x=\boldsymbol{e}_r\sin\theta\cos\phi+\boldsymbol{e}_\theta\cos\theta\cos\phi-\boldsymbol{e}_\phi\sin\phi$ $\boldsymbol{e}_y=\boldsymbol{e}_r\sin\theta\sin\phi+\boldsymbol{e}_\theta\cos\theta\sin\phi-\boldsymbol{e}_\phi\cos\phi$ $\boldsymbol{e}_z=\boldsymbol{e}_r\cos\theta-\boldsymbol{e}_\theta\sin\theta$
在球坐标系中矢量表示	$\boldsymbol{a}(\boldsymbol{l})=\boldsymbol{a}(r,\theta,\phi)=\boldsymbol{e}_r a_r(r,\theta,\phi)+\boldsymbol{e}_\theta a_\theta(r,\theta,\phi)+\boldsymbol{e}_\phi a_\phi(r,\theta,\phi)$
微分长度元	$\mathrm{d}\boldsymbol{l}=\boldsymbol{e}_r\mathrm{d}r+\boldsymbol{e}_\theta r\mathrm{d}\theta+\boldsymbol{e}_\phi r\sin\theta\mathrm{d}\phi$
体积元	$\mathrm{d}V=r^2\sin\theta\mathrm{d}r\mathrm{d}\theta\mathrm{d}\phi$
面积元	$\mathrm{d}S_r=r^2\sin\theta\mathrm{d}\theta\mathrm{d}\phi,\mathrm{d}S_\theta=r\sin\theta\mathrm{d}r\mathrm{d}\phi,\mathrm{d}S_\phi=r\mathrm{d}r\mathrm{d}\theta,$ $\mathrm{d}\boldsymbol{S}=\boldsymbol{e}_r\mathrm{d}S_r+\boldsymbol{e}_\theta\mathrm{d}S_\theta+\boldsymbol{e}_\phi\mathrm{d}S_\phi$
哈密顿算符	$\nabla=\boldsymbol{e}_r\frac{\partial}{\partial r}+\boldsymbol{e}_\theta\frac{1}{r}\frac{\partial}{\partial\theta}+\boldsymbol{e}_\phi\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}$
梯度	$\nabla\varphi=\boldsymbol{e}_r\frac{\partial\varphi}{\partial r}+\frac{\boldsymbol{e}_\theta}{r}\frac{\partial\varphi}{\partial\theta}+\frac{e_\phi}{r\sin\theta}\frac{\partial\varphi}{\partial\phi}$
散度	$\nabla\cdot\boldsymbol{a}=\frac{1}{r^2}\frac{\partial}{\partial r}(r^2a_r)+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta a_\theta)+\frac{1}{r\sin\theta}\frac{\partial a_\phi}{\partial\phi}$

旋度	$$\nabla\times\boldsymbol{a}=\begin{vmatrix}\frac{1}{r^2\sin\theta}\boldsymbol{e}_r & \frac{1}{r\sin\theta}\boldsymbol{e}_\theta & \frac{1}{r}\boldsymbol{e}_\phi\\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi}\\ a_r & ra_\theta & r\sin\theta a_\phi\end{vmatrix}$$ $$=\boldsymbol{e}_r\left(\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta a_\phi)-\frac{1}{r\sin\theta}\frac{\partial a_\theta}{\partial\phi}\right)$$ $$+\boldsymbol{e}_\theta\left(\frac{1}{r\sin\theta}\frac{\partial a_r}{\partial\phi}-\frac{1}{r}\frac{\partial}{\partial r}(ra_\phi)\right)+\boldsymbol{e}_\phi\left(\frac{1}{r}\frac{\partial}{\partial r}(ra_\theta)-\frac{1}{r}\frac{\partial a_r}{\partial\theta}\right)$$
拉普拉斯运算	$$\nabla^2\varphi=\Delta\varphi=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\varphi}{\varphi r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial\varphi}{\partial\theta})+\frac{1}{r^2\sin^2\theta}\frac{\partial^2\varphi}{\partial\phi^2}$$

2 常用矢量公式

$$\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c})=\boldsymbol{b}\cdot(\boldsymbol{c}\times\boldsymbol{a})=\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b})$$

$$\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})=(\boldsymbol{a}\cdot\boldsymbol{c})\boldsymbol{b}-(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c}$$

$$\nabla\cdot\nabla=\nabla^2=\Delta$$

$$\nabla(\varphi_1\varphi_2)=\varphi_1\nabla\varphi_2+\varphi_2\nabla\varphi_1$$

$$\nabla\cdot(\varphi\boldsymbol{a})=\boldsymbol{a}\cdot\nabla\varphi+\varphi\nabla\cdot\boldsymbol{a}$$

$$\nabla\times(\varphi\boldsymbol{a})=\nabla\varphi\times\boldsymbol{a}+\varphi\nabla\times\boldsymbol{a}$$

$$\nabla(\boldsymbol{a}\cdot\boldsymbol{b})=(\boldsymbol{a}\cdot\nabla)\boldsymbol{b}+(\boldsymbol{b}\cdot\nabla)\boldsymbol{a}+\boldsymbol{a}\times(\nabla\times\boldsymbol{b})+\boldsymbol{b}\times(\nabla\times\boldsymbol{a})$$

$$\nabla\cdot(\boldsymbol{a}\times\boldsymbol{b})=\boldsymbol{b}\cdot\nabla\times\boldsymbol{a}-\boldsymbol{a}\cdot\nabla\times\boldsymbol{b}$$

$$\nabla\times(\boldsymbol{a}\times\boldsymbol{b})=\boldsymbol{a}\nabla\cdot\boldsymbol{b}-\boldsymbol{b}\nabla\cdot\boldsymbol{a}+(\boldsymbol{b}\cdot\nabla)\boldsymbol{a}-(\boldsymbol{a}\cdot\nabla)\boldsymbol{b}$$

$$\nabla\cdot(\nabla\varphi)=(\nabla\cdot\nabla)\varphi=\nabla^2\varphi=\Delta\varphi$$

$$\nabla\times(\nabla\varphi)=0$$

$$\nabla\cdot(\nabla\times\boldsymbol{a})=0$$

$$\nabla\times(\nabla\times\boldsymbol{a})=\nabla(\nabla\cdot\boldsymbol{a})-\nabla^2\boldsymbol{a}$$

$$\oint_S\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S}=\int_V\nabla\cdot\boldsymbol{a}\mathrm{d}V$$

$$\oint_C\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{l}=\int_s\nabla\times\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{S}$$

$$\oint_S(\boldsymbol{n}\times\boldsymbol{a})\mathrm{d}S=\int_V\nabla\times\boldsymbol{a}\cdot\mathrm{d}V$$

$$\oint_S\varphi\boldsymbol{n}\mathrm{d}S=\int_V\nabla\varphi\mathrm{d}V$$

$$\oint_C\varphi\mathrm{d}\boldsymbol{l}=\int_S\boldsymbol{n}\times\nabla\varphi\mathrm{d}S$$